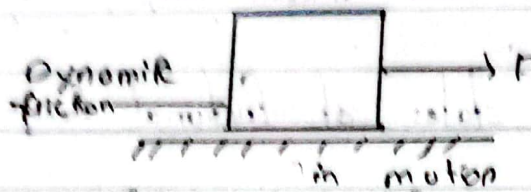


Dynamic friction

- Acts when the body is in movement
- It opposes the motion and acts in the opposite direction to the motion.



Factors that affect friction

1. Nature of the surface
2. Area of contact
3. Mass

Practical applications of friction

- 1) Various parts of most of the machines and instruments that we use daily are in contact with other parts. When we operate these machines and instruments, these parts slide on one another, giving rise to frictional forces.

Therefore when machines are operated, an excessive amount of additional work has to be done against frictional forces, causing a loss of energy. This loss of energy appears as heat, raising the temperature of the objects. If we could reduce frictional forces, then we would be able to minimize the energy loss and the rise in temperature.

2. Find examples from your home environment where friction is useful and unwanted. Copy down the table below and complete it with at least three examples for each category.

(A) Useful applications of friction.

Example	How friction is useful.
Walking on floor	Friction between your feet (or shoes) and the floor prevents you from slipping and allows you
Gripping a glass or pen	Friction between your fingers and the object allows you to hold it securely without it sliding out of your hand.
Striking a matchstick	Rubbing a matchstick against the rough surface of the box creates the heat necessary to ignite the match

(B) Unwanted applications of friction

Example	Why friction is unwanted
1. Squeaky door hinges	Friction between the metal parts of the hinge causes resistance and an annoying noise.
2. Moving heavy furniture	Friction between the furniture legs and the floor makes it difficult to push and can scratch the surface of the floor
3. Soles of shoes wearing out	Constant friction with the ground gradually wears away the material, eventually making the shoes unusable or slippery.